



POLITECNICO
MILANO 1863

Dynamics of Electrical Machines and Drives

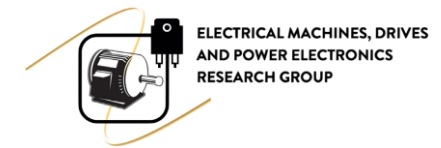
Introduction

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Electrical machines, Drives and power electronics research group

□ Location:

Department of Mechanical Engineering
Building B23, 1st floor
Office: Prof. Francesco Castelli Dezza
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□ Appointments for questions & explanations: [send an email](#)

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Practical Lessons Overview

Excercises

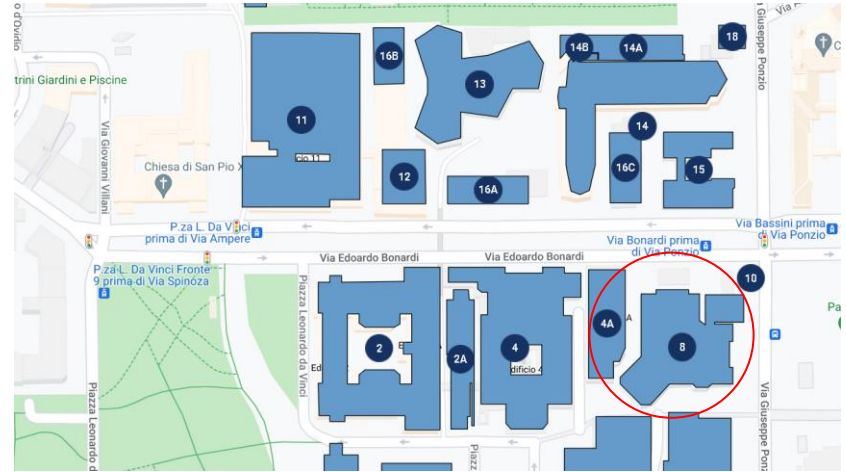
Practical lectures will take place at:

❑ [Aula 8.o.1](#)
Building 8
Campus Leonardo
Via Bonardi, 9 - 20133 - Milano (MI)

Tuesday 16:15 – 19:00

❑ course pages links:

❑ Students are required to bring their own PCs



Practical Lectures Overview

Exercises

- ❑ The exercises relates to using computer simulations tools (MATLAB/Simulink) to:
 - analyze the behavior of a dynamic system
 - analyze the effect of the control loop on the performance of an electrical drive

Final exam

The exam consists of an oral test which verify:

- the knowledge of the control of the electric drives (Prof. Castelli Dezza lectures)
- and a discussion on the report provided by the student (provide a report is mandatory)



during the practical lectures, the students will encounter both:

- **Standard exercises** (not suitable for the report)
- **Report exercises** (suitable for the report)
 - ✓ there will be four of this exercises during the semester
 - ✓ the students can pick one of them and prepare the report (at any time)
 - ✓ the report has to show the definition of the parameters of the control system and the results/figures of the Simulink simulation

NB exercises can be done as groups... BUT each student MUST present its own report

Suggested Book

Exercises and Embedded Implementation

- ❑ The exercises presented in this course are extensively detailed in the book:



[Introduction to Microcontroller Programming for Power Electronics Control Applications: Coding with MATLAB® and Simulink® \(1st ed.\)](#)

CRC Press, 2021

<https://doi.org/10.1201/9781003196938>

M. Rossi, N. Toscani, M. Mauri, and F. Castelli-Dezza

This book also cover the related *embedded implementation aspects on MCUs* which are not covered in this course.

Available in the campus library at:

<https://ebookcentral.proquest.com/lib/polimi/detail.action?docID=6710080>

Definition:

A dynamical system describes the evolution of a state over time.

We need to specify what we mean for “evolution”, “state” and “time”:

① *continuous time* $t \in \mathbb{R}$

the evolution of the state is described by *ordinary differential equations* (ODE). Think of the linear, continuous time system in state space form

$$\dot{x}(t) = Ax(t)$$

② *discrete time* $k \in \mathbb{Z}$

the evolution of the state is described by a difference equation. Think of the linear discrete time system in state space form

$$x(k+1) = Ax(k)$$

Basics for Electrical Drives Simulation

Evolution of the State

- ① We want to **observe** and control the currents $x(t) = i(t) \rightarrow y(t) = x(t)$
- ② Voltage $u(t) = v(t)$ is a **forcing signal**
- ③ Adding the information about **initial conditions** $x_0 \in \mathbb{R}^n$

$$\begin{cases} \dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t) \\ x(0) &= x_0 \end{cases}$$

where

$$A = -\frac{R}{L} \quad B = \frac{1}{L} \quad C = 1 \quad D = 0$$

- the **evolution of the state** $x(t) = y(t)$ is

$$x(t) = \underbrace{e^{At} x_0}_{\substack{\text{natural response} \\ \text{(effect of initial condition)}}} + \underbrace{\int_0^t e^{A(t-\tau)} B u(\tau) d\tau}_{\substack{\text{forced response} \\ \text{(effect of input signal)}}}$$

- if $A < 0 \Rightarrow$ **asymptotically stable**

Basics for Electrical Drives Simulation

Transfer function

Laplace transforms convert integral and differential equations into algebraic equations through the Laplace operator

$$F(s) = \mathcal{L}[f(t)] \quad \leftarrow \quad s = d/dt$$

It can be demonstrated that:

$$y(s) = \underbrace{C(sI - A)^{-1}x_0}_{\text{Laplace transform of natural response}} + \underbrace{(C(sI - A)^{-1}B + D)u(s)}_{\text{Laplace transform of forced response}}$$

Definition:

The transfer function $G(s)$ of a continuous-time linear system (A, B, C, D) is $G(s) = C(sI - A)^{-1}B + D$ between the Laplace transform $y(s)$ of output and the Laplace transform $u(s)$ of the input signals for the initial state $x_0 = 0$

Basics for Electrical Drives Simulation

Current Control of an RL load

- electrical machines windings behaves as **RL (ohmic-inductive)** load
- controlling the *current flow* into windings imply *control the torque*

Kirchhoff's voltage law :
$$v(t) - L \frac{di(t)}{dt} - Ri(t) = 0$$

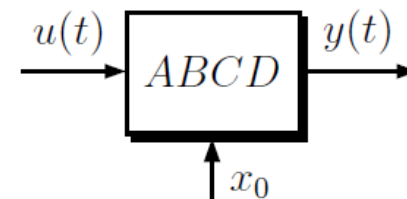
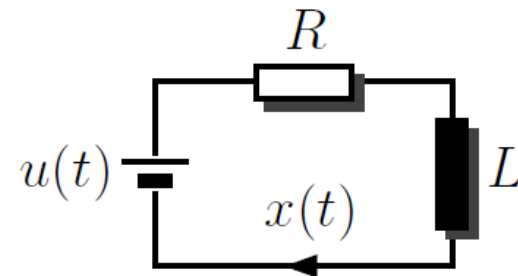
- 1 Rewrite it as a 1st order linear system

$$\frac{di(t)}{dt} = \frac{1}{L}v(t) - \frac{R}{L}i(t)$$

- 1 or in a **state-space form**

$$\dot{x}(t) = \underbrace{-R/L}_A x(t) + \underbrace{1/L}_B u(t)$$

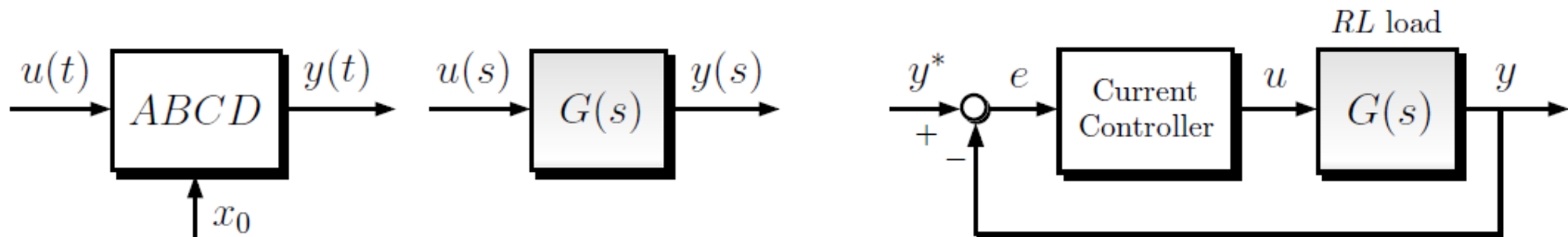
where $x(t) = i(t)$ and $u(t) = v(t)$



Basics for Electrical Drives Simulation

Current Control (PI-based) of an RL Load

Note that $y(s) = i(s)$ and $u(s) = v(s)$



The RL load to be controlled has a time constant $TG = L/R$

The controller settings are chosen according to:

- ① Time response $\rightarrow$ Bandwidth ω_c
- ② Robustness level $\rightarrow$ Phase margin φ_m
- ③ Sensitivity to disturbances (and actuations) $\rightarrow S(s)$

PI-Action

No pole-zero cancellation

Therefore, $F(j\omega)$ can be explicated as:

$$|F(j\omega_c)|^2 = \frac{1}{2} \rightarrow k_i^2 + 2\omega_c^2 k_i L + \omega_c^2 k_p^2 - \omega_c^2 (R^2 + 2k_p R) - \omega_c^4 L^2 = 0$$

The solution of this equation leads to:

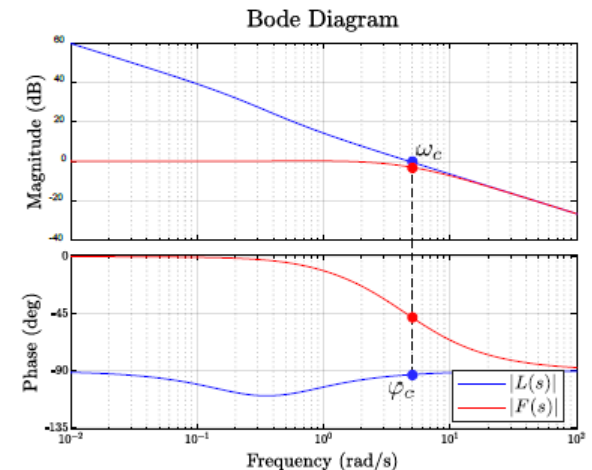
$$k_i = -\omega_c^2 L \pm \omega_c \sqrt{2\omega_c^2 L^2 - k_p^2 + R^2 + 2k_p R}$$

k_i must be $k_i > 0$, $k_i \in \mathbb{R}$. Therefore:

$$R - \sqrt{2R^2 + \omega_c^2 L^2} < k_p < \underbrace{R + \sqrt{2R^2 + \omega_c^2 L^2}}_{k_p \text{ max}}$$

A **practical rule** is to set $k_p = 0.9k_{p,\text{max}}$. Consequently:

$$\varphi_m = \pi - |\angle L(j\omega_c)| \rightarrow 90^\circ \quad \angle L(j\omega_c) = \frac{\pi}{2} - \arctan\left(\frac{k_i}{\omega_c k_p}\right) - \arctan\left(\frac{L}{R}\right)$$



PI-Action

Pole-zero cancellation

Alternatively, it is possible to consider the following controller expression: $R(s) = k_p (1 + sT_i) / sT_i$ where $T_i = k_p/k_i$

Therefore:

$$L(s) = G(s)R(s) = k_p \frac{1 + sT_i}{sT_i} \frac{1}{R + sL} = \frac{k_p}{R} \frac{\cancel{1 + sT_i}}{sT_i} \frac{1}{\cancel{1 + sL/R}}$$

It is possible to perform a **pole/zero cancellation** imposing:

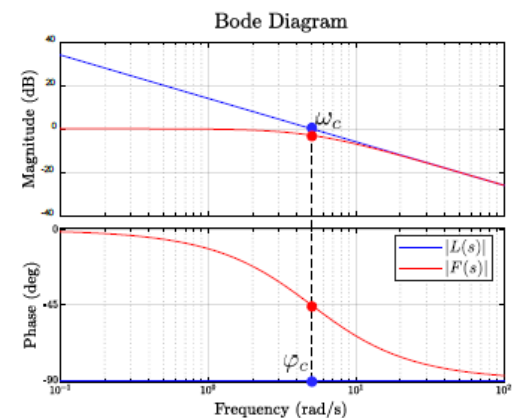
$$T_i = L/R \quad \begin{aligned} k_p &= \omega_c L \rightarrow k_p/k_i = L/R \\ k_i &= \omega_c R \end{aligned}$$

Thus:

$$L(s) = \frac{k_p}{sRT_i}$$

$$|L(j\omega_c)| = 1 \rightarrow \omega_c = \frac{k_p}{RT_i} = \frac{k_i}{R}$$

Different behaviour
than before



Exercise

Current Control of an RL load on Matlab

Let's assume an $L=0.1\text{H}$ and $R=0.025$

The transient has a time constant $\tau = L/R$ and this means that the constant value will be reached at $T = 5\tau$.

I want a controller that is capable of reaching the transient end at $T_c=1\text{s}$.

How can I do it on Matlab?

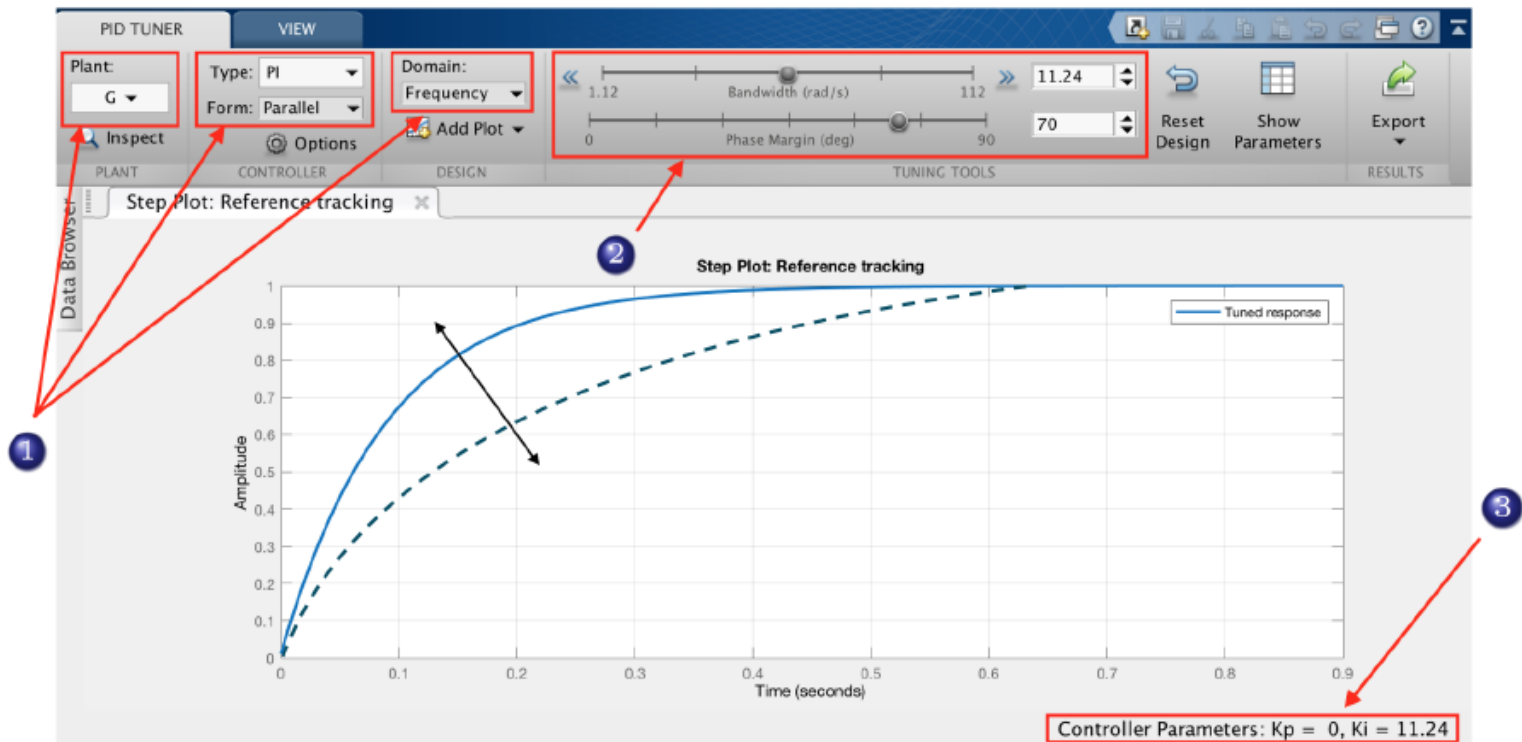
Try the following MATLAB functions:

- `tf()`
- `step()`
- `bode()`
- `margin()`
- `piddtune()`

The PID-tuner

Direct approach with MATLAB-pidtool

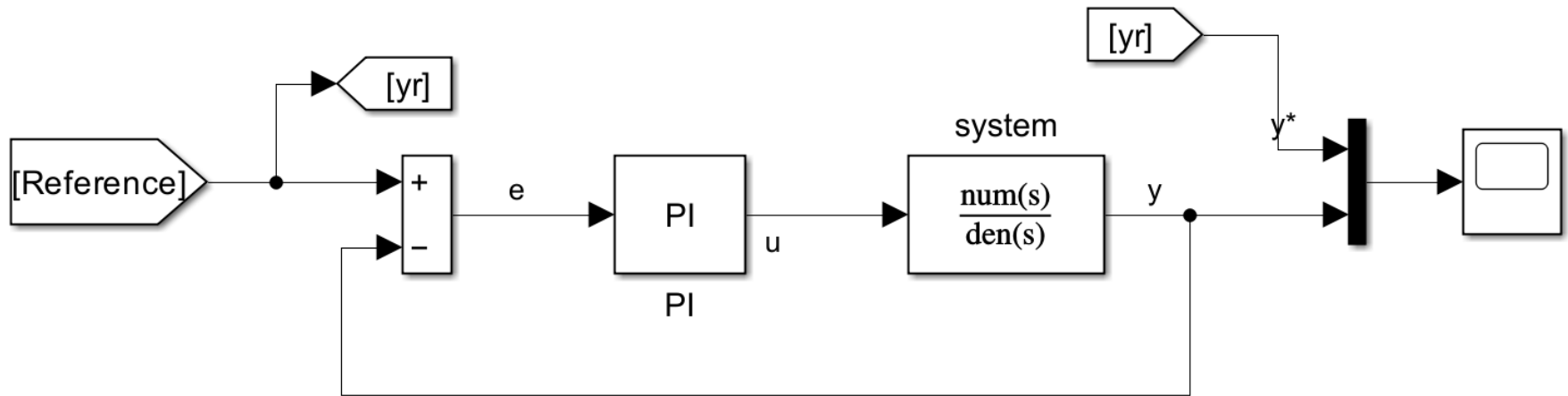
Given the plant $G(s)$ this interface allow you to select the controller type and directly see the effects of parameter changing.



Exercise

Current Control of an RL load on Simulink

Now we have a complete modelled system. Let's try to put it on Simulink.



How is my system going to react to different inputs? Do I have to change the regulators parameters? Which ones?

Try several inputs: step, sinusoidal, ramp.